speaker architecture found within the particular vehicle into which the radio has been installed.

Proper testing of the PCI data bus and the electronic data bus message inputs to and outputs from the EMIC that control audio system cabin equalization requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

AXLE LOCKER CONTROL

The EMIC contains a logic circuit and programming to perform the axle locker control functions for models equipped with the optional off-road package. The EMIC monitors hard wired inputs from the ignition switch on the fused ignition switch output (run-start) circuit and the key-in ignition switch on the key-in ignition switch sense circuit. The EMIC also monitors vehicle speed and transfer case shift position switch electronic message inputs received from the Powertrain Control Module (PCM) over the PCI data bus. The internal programming of the EMIC then determines whether to activate or deactivate the axle locker function by enabling or disabling the axle lock switch located in the instrument panel accessory switch bezel.

The EMIC programming controls the axle lock switch through two separate axle lock switch enable circuits, enable 1 and enable 2. In all cases, the EMIC will not activate either enable circuit if there are any transfer case shift position switch or vehicle speed sensor faults present. Whenever the ignition switch is in the On or Start positions, the key is in the ignition lock cylinder, the transfer case is in 4 X 4 Low range, and the vehicle speed is less than about 72 kilometers per hour (45 miles per hour) the first enable (enable 1) circuit is activated. The second enable (enable 2) circuit is activated only if the vehicle speed is less than about 16 kilometers per hour (10 miles per hour). When both enable circuits are activated, the axle lock switch becomes functional.

Once activated, the enable 1 circuit is automatically deactivated whenever the transfer case is moved out of the 4 X 4 Low range, or if the vehicle speed is greater than about 72 kilometers per hour (45 miles per hour). If the enable 1 circuit is deactivated after the rear or the front and rear axle lockers are engaged, all outputs from the axle lock switch are dropped causing both axles to unlock. The enable 2 circuit is automatically deactivated whenever the vehicle speed is greater than about 16 kilometers per hour (10 miles per hour). If the enable 2 circuit is deactivated after the rear or the front and rear axle lockers are engaged, the outputs from the axle lock switch are unaffected and the locked axles remain locked. However, an unlocked axle cannot be locked until the vehicle speed is reduced and the enable 2 circuit is again activated.

In addition, once activated, both enable circuits will remain active regardless of the status of the ignition switch input. Therefore, any locked axle will remain locked and the various components of the axle locker system will remain functional after the ignition switch is turned to the Off position. However, while the currently selected axle locker mode remains active with the ignition switch turned Off, if the key is removed from the ignition lock cylinder, Off is the only other axle locker mode that can be selected with the axle lock switch. For as long as the key is removed from the ignition lock cylinder, the cluster logic will interpret any revision to the input status of either request circuit from the axle lock switch as a cancellation request and will deactivate the enable 1 circuit and all outputs from the axle lock switch are dropped, causing both axles to unlock. Otherwise, once locked, any locked axle will remain locked until the axle lock switch is deactivated (enable 1 circuit is deactivated), or until the Off mode is manually selected by moving the axle lock switch rocker to the Off position. The EMIC also provides the vehicle operator with distinct visual and/or audible indications as to the current status of the axle locker system through chime warnings and illumination of the rear/front lock indicators in the cluster as outlined elsewhere in this service information.

The hard wired input and output circuits of the EMIC axle locker control may be diagnosed using conventional diagnostic methods. However, proper testing of the EMIC programming and the electronic vehicle speed and transfer case shift position switch messages received by the EMIC over the PCI data bus requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

CHIME SERVICE

The EMIC is equipped with hardware and software to provide chime service for all available features in the chime warning system. Upon receiving the proper chime inputs, the EMIC activates an integral on-board audible tone generator to provide audible chime tones to the vehicle operator. The chime tone generator in the EMIC is capable of producing single chime tones or repeated chime tones at two different rates: a slow rate of about fifty chime tones per minute, and a fast rate of about 180 chime tones per minute. The internal programming of the EMIC determines the priority of each chime tone request input that is received, as well as the rate and duration of each chime tone that is to be generated.

The EMIC relies upon hard wired inputs from the door ajar switches, the left multi-function switch, the ignition switch, and the park brake/brake warning indicator switches to provide chime service for the